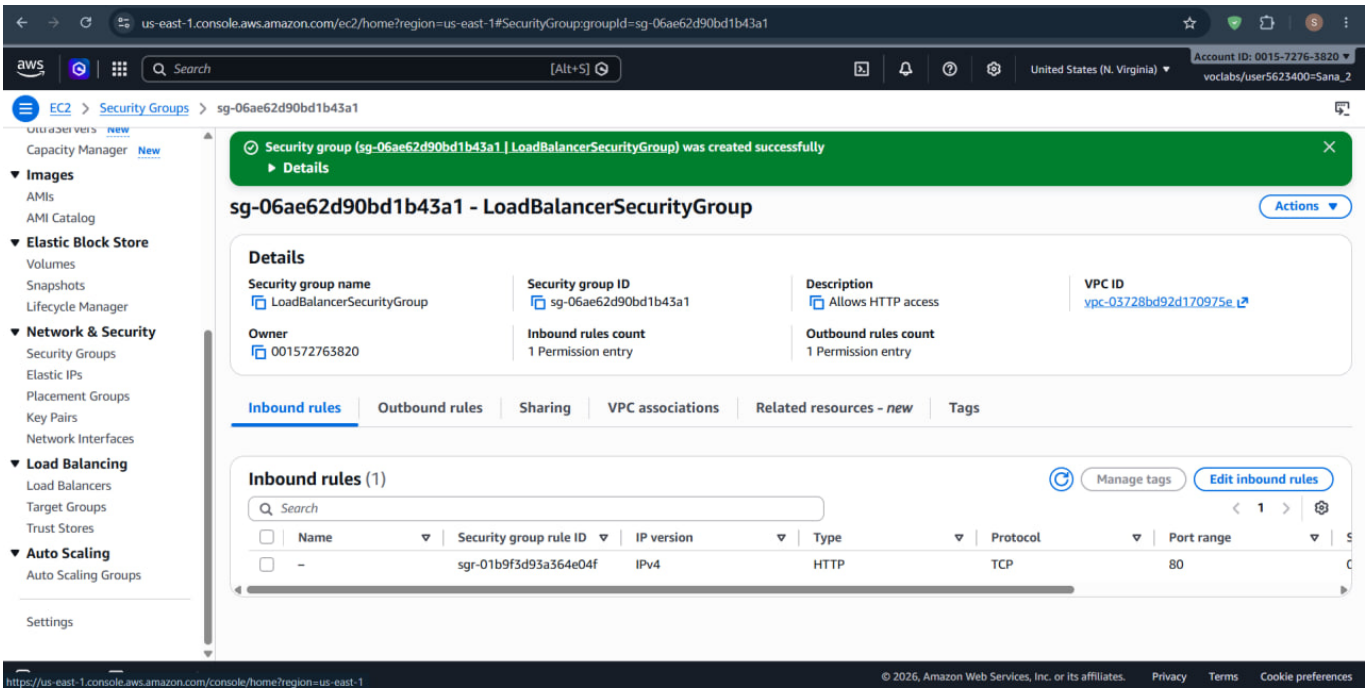
Cloud Computing - Lab 2: Elastic Load Balancing & Auto Scaling

- **Student ID**: e20220671
- **Course**: Cloud Computing

1. Security Groups Configuration

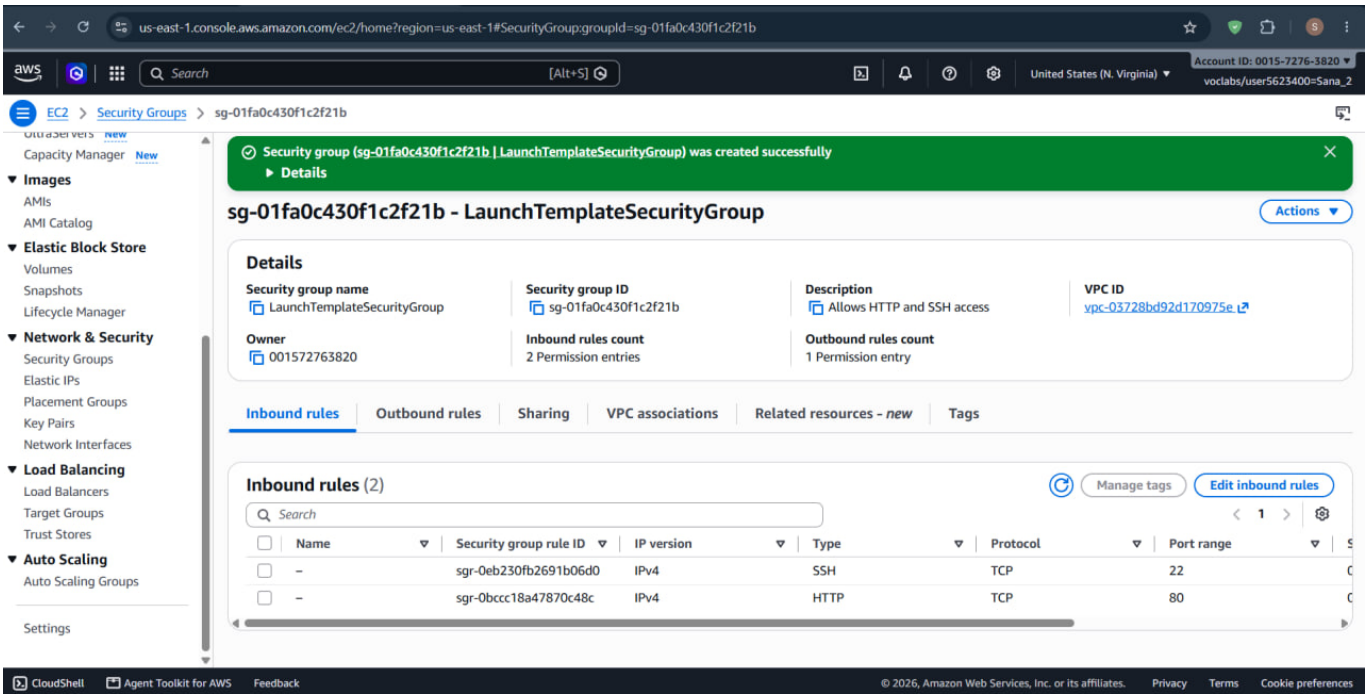
1.1 Load Balancer Security Group

Created `LoadBalancerSecurityGroup` (`sg-06ae62d90bd1b43a1`) to allow inbound HTTP (port 80) traffic to the Application Load Balancer.



1.2 Launch Template Security Group

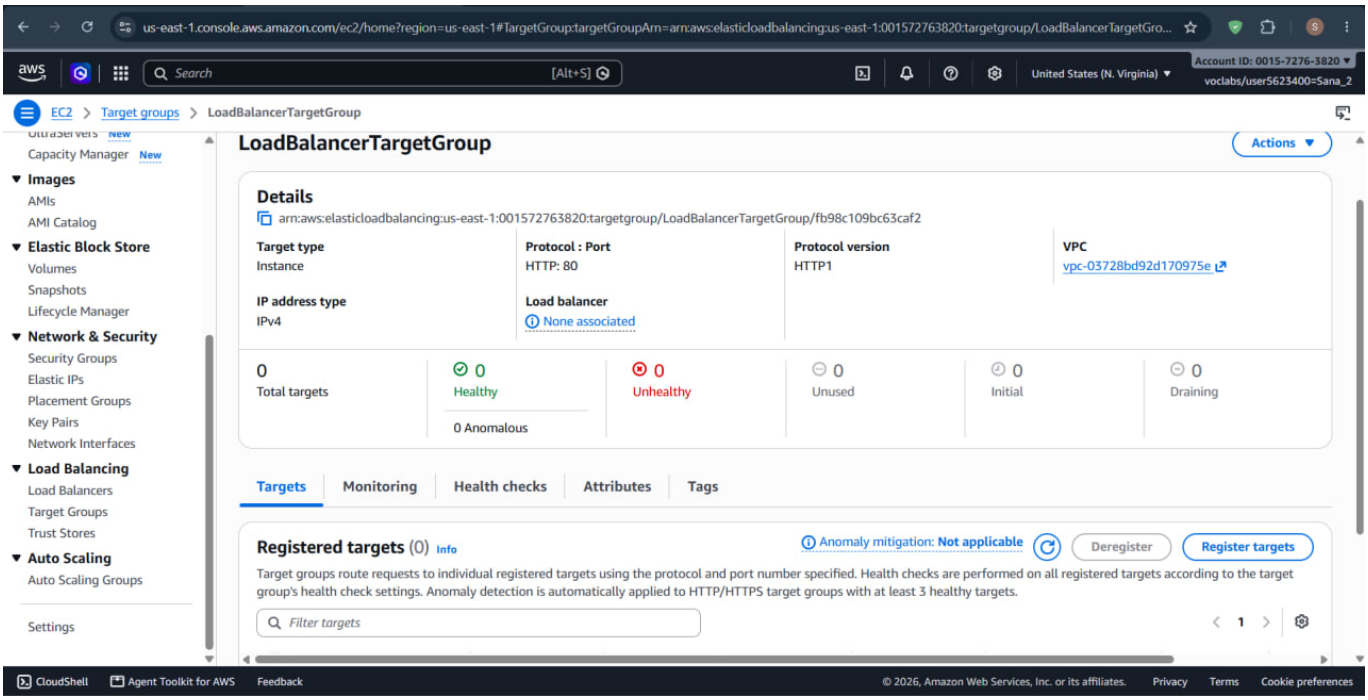
Created `LaunchTemplateSecurityGroup` (`sg-01fa0c430f1c2f21b`) to allow inbound HTTP (port 80) and SSH (port 22) access to the EC2 instances.



2. Target Group Configuration

2.1 Target Group Details

Created target group `LoadBalancerTargetGroup` with target type `Instance` on `HTTP: 80` (IPv4).

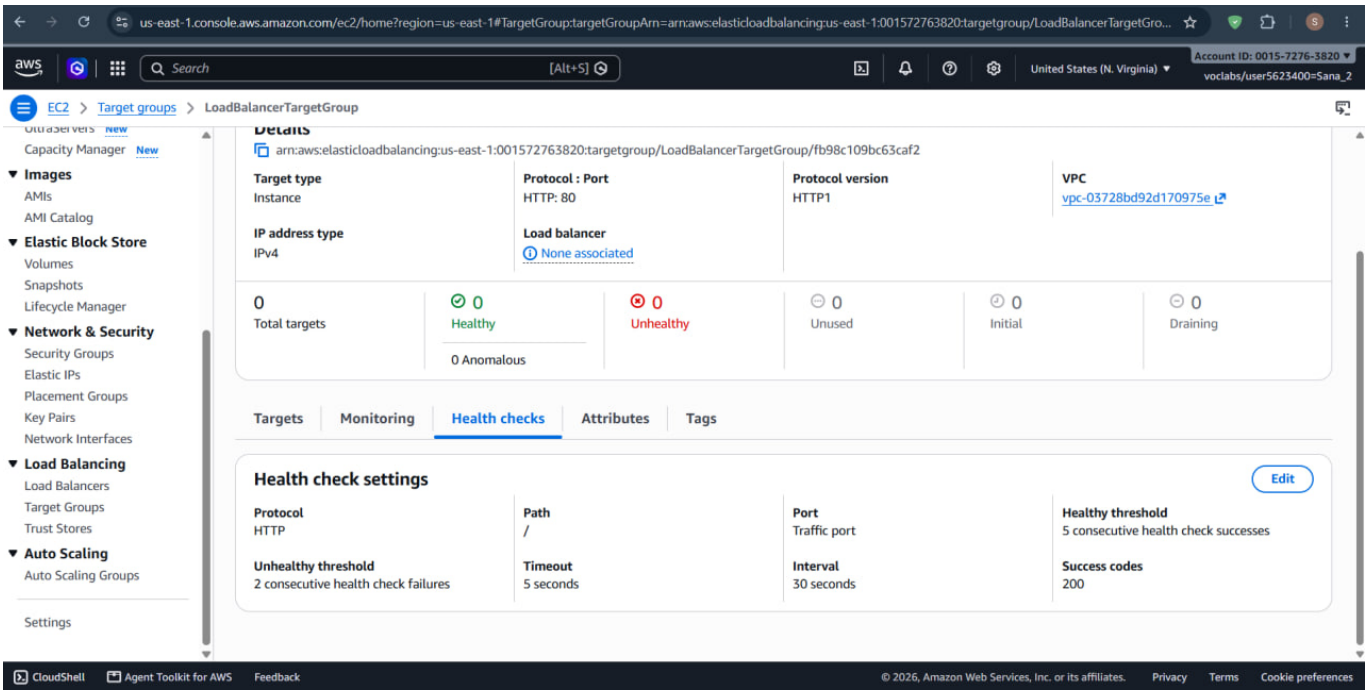


2.2 Health Checks Configuration

Configured health check settings on `LoadBalancerTargetGroup`:

- **Protocol**: `HTTP`
- **Path**: `/`
- **Healthy threshold**: `5` consecutive health check successes
- **Unhealthy threshold**: `2` consecutive health check failures

- **Timeout**: `5 seconds`
- **Interval**: `30 seconds`
- **Success codes**: `200`

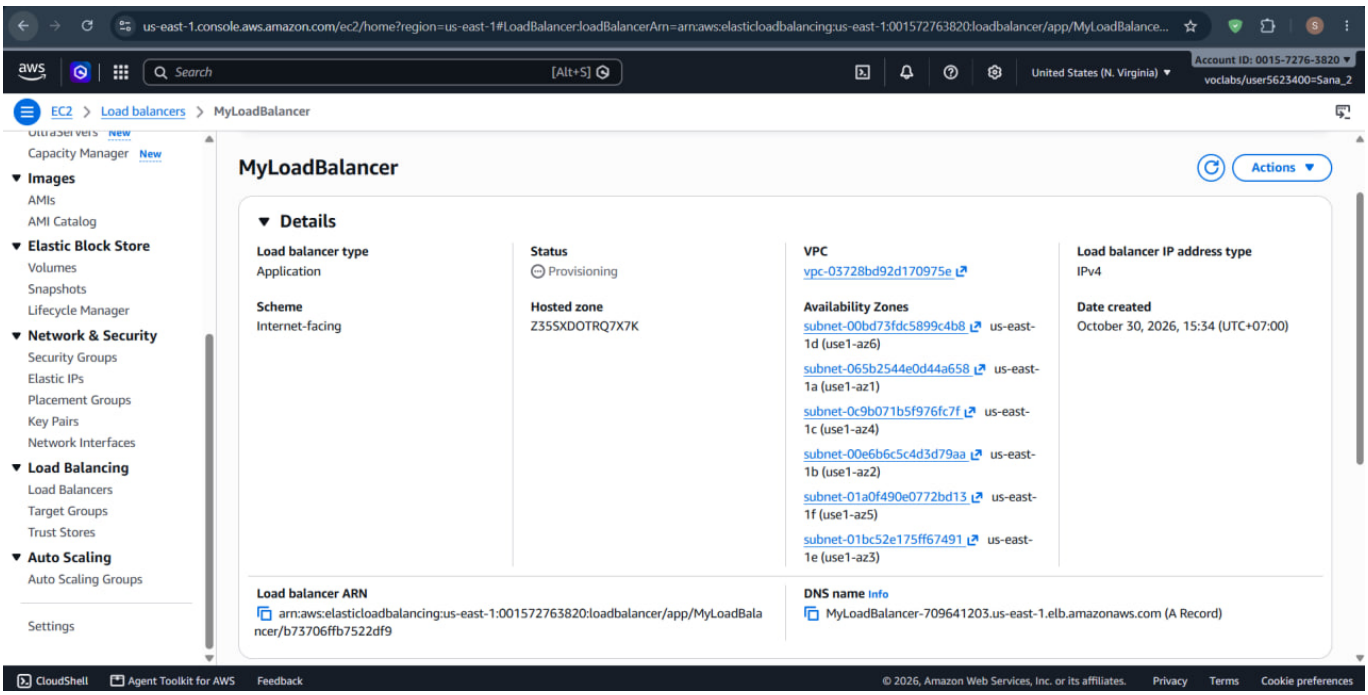


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3. Application Load Balancer (ALB)

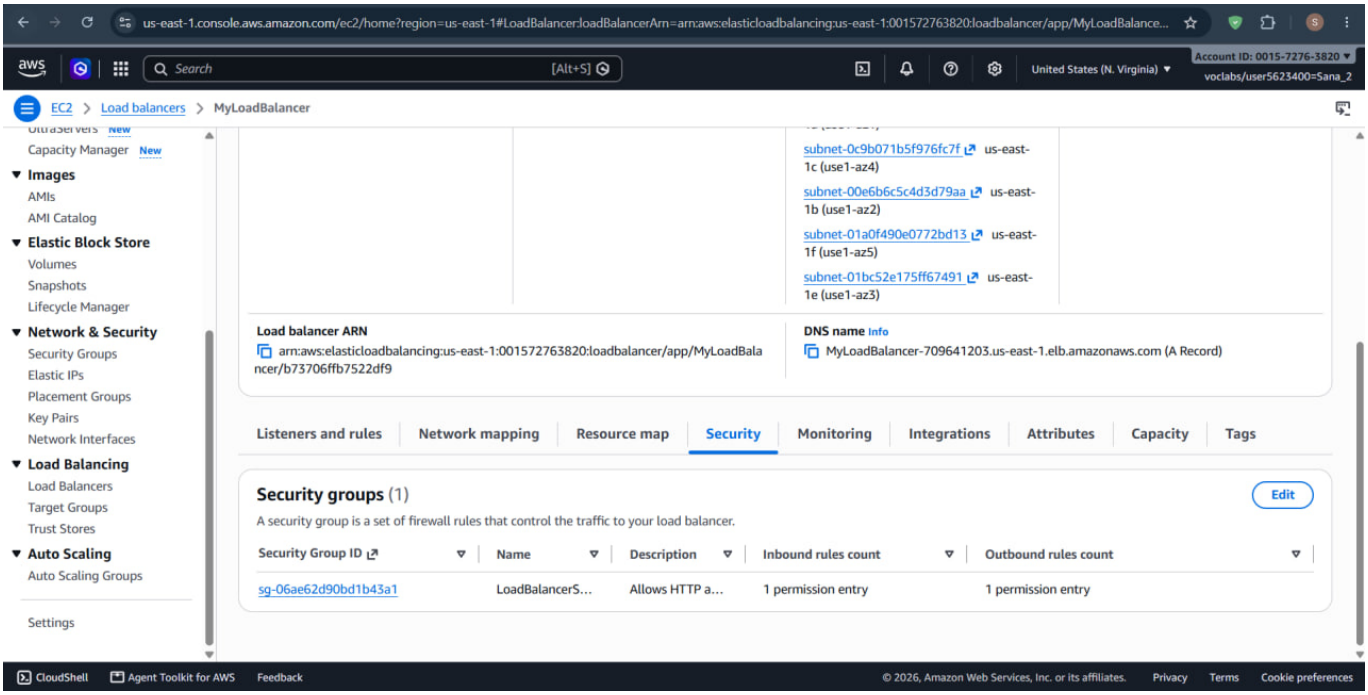
3.1 Load Balancer Overview

Provisioned internet-facing Application Load Balancer `MyLoadBalancer` (`MyLoadBalancer-709641203.us-east-1.elb.amazonaws.com`) across the VPC availability zones.



3.2 Load Balancer Security & Details

Detailed view of `MyLoadBalancer` showing the attached `LoadBalancerSecurityGroup` (`sg-06ae62d90bd1b43a1`).



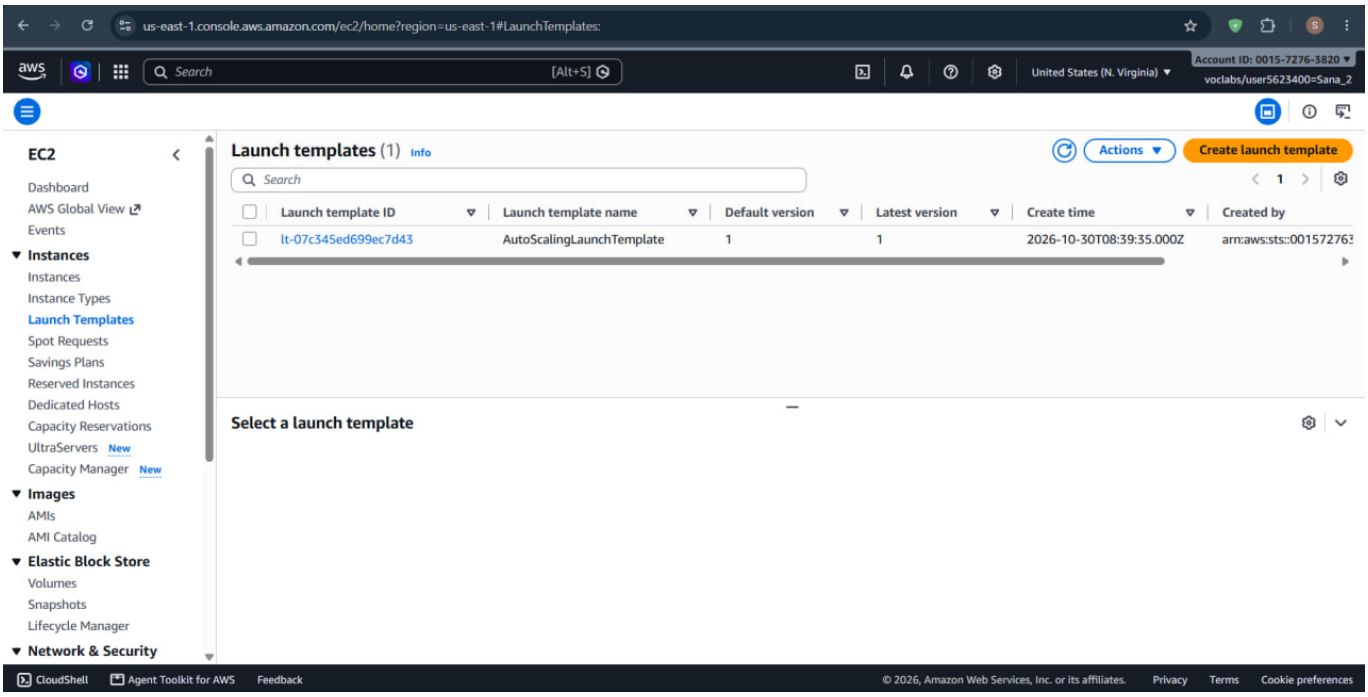
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4. Launch Template & Auto Scaling Group (ASG)

4.1 Launch Template

Created launch template `AutoScalingLaunchTemplate` (`lt-07c345ed699ec7d43`) configured with Amazon Linux 2023 AMI (`ami-073f45061d35c86f0`), instance type `t2.micro`, security group `LaunchTemplateSecurityGroup`, and user data script to bootstrap Apache (`httpd`):

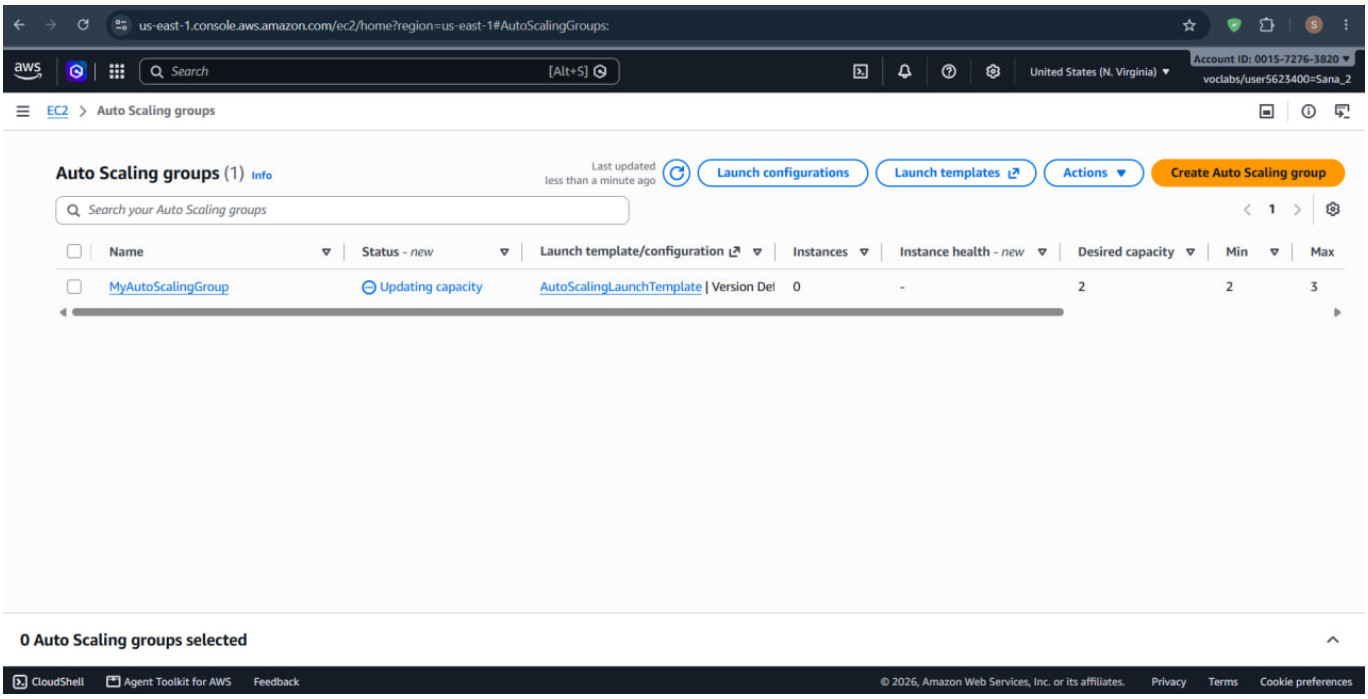
```
#!/bin/bash
dnf update -y
dnf install -y httpd
systemctl start httpd
systemctl enable httpd
echo "Hello World from ($(hostname -f))" > /var/www/html/index.html
```



4.2 Auto Scaling Group List

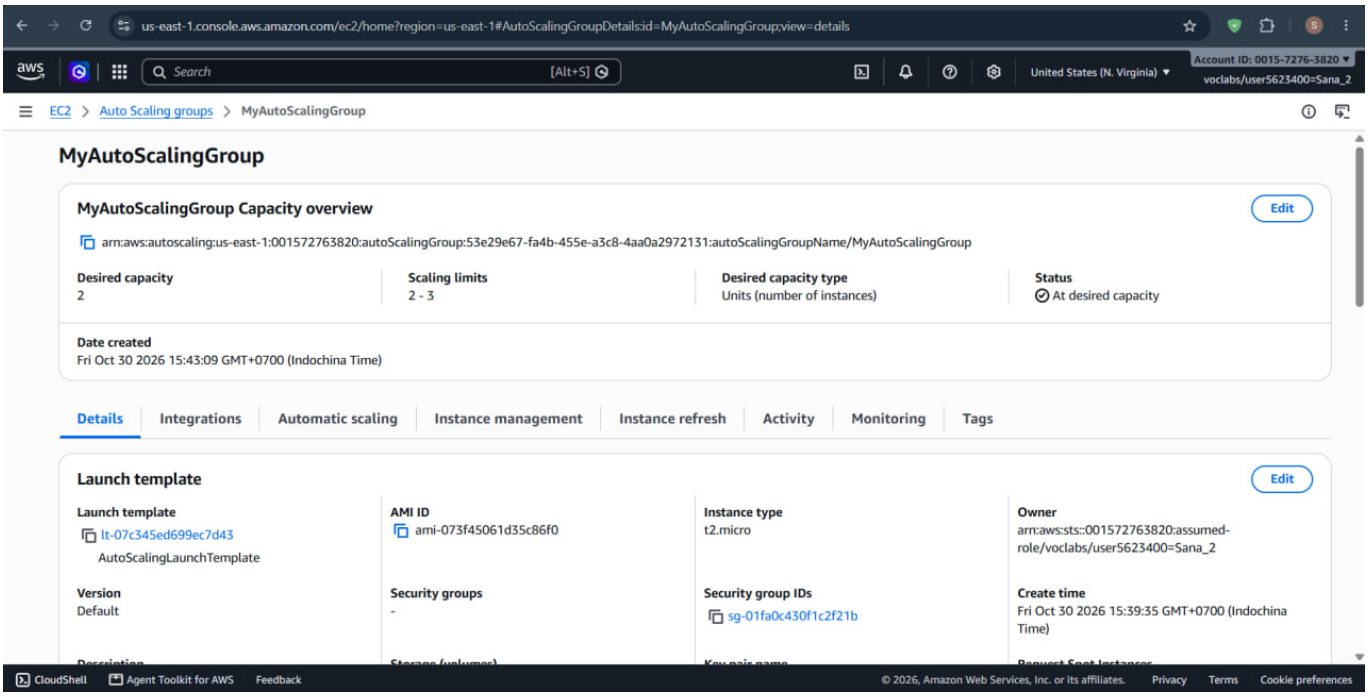
Created Auto Scaling Group `MyAutoScalingGroup` using `AutoScalingLaunchTemplate` with:

- **Desired capacity**: 2
- **Min capacity**: 2
- **Max capacity**: 3



4.3 Auto Scaling Group Details

Detailed configuration of `MyAutoScalingGroup` showing capacity status (`At desired capacity`) and associated launch template details.



5. Load Balancing Verification

5.1 Browser Test — Instance 1

Accessing the Load Balancer DNS URL (`http://myloadbalancer-709641203.us-east-1.elb.amazonaws.com`), returning response from the first EC2 instance (`ip-172-31-94-96.ec2.internal`):



Hello World from (ip-172-31-94-96.ec2.internal)

5.2 Browser Test — Instance 2

Refreshing the Load Balancer DNS URL, demonstrating round-robin traffic distribution to the second EC2 instance (`ip-172-31-0-135.ec2.internal`):



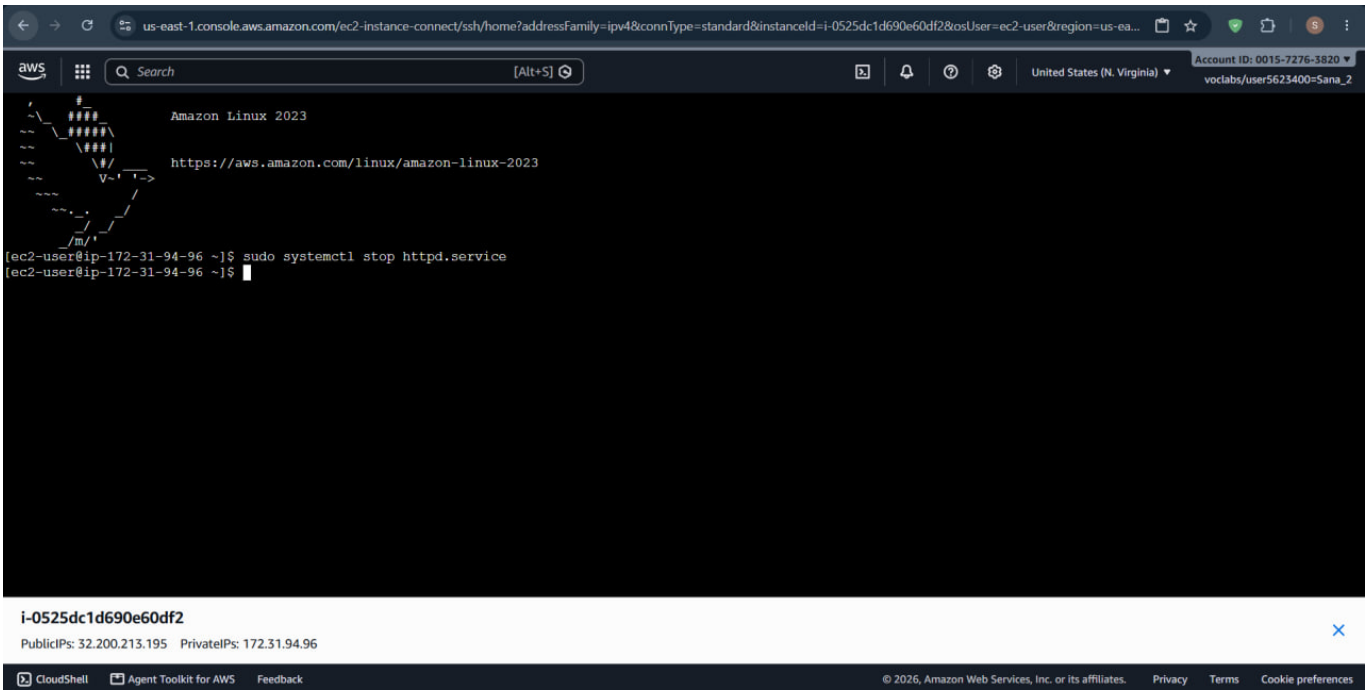
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6. High Availability & Self-Healing Test

6.1 Stopping Apache Service (`httpd`)

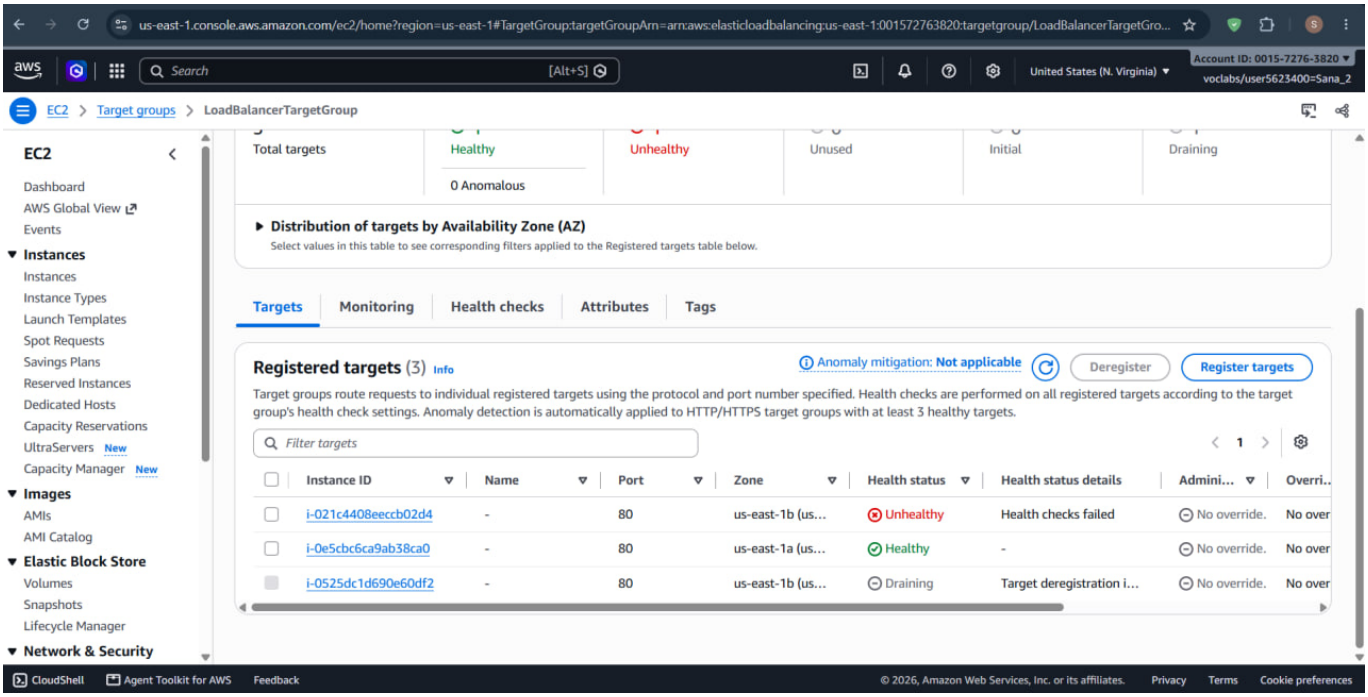
Connected to instance `i-0525dc1d690e60df2` (`ip-172-31-94-96`) via EC2 Instance Connect and stopped the Apache service to simulate a failure:

```
sudo systemctl stop httpd.service
```



6.2 Unhealthy Target Detection

Target group `LoadBalancerTargetGroup` health checks detecting the failure (`Health checks failed` / `Draining` during target deregistration).



6.3 Automatic Instance Replacement

Auto Scaling Group automatically replacing the failed instance and restoring `LoadBalancerTargetGroup` to `2` healthy targets (`i-021c4408eeccb02d4` and `i-0e5cbc6ca9ab38ca0`).

